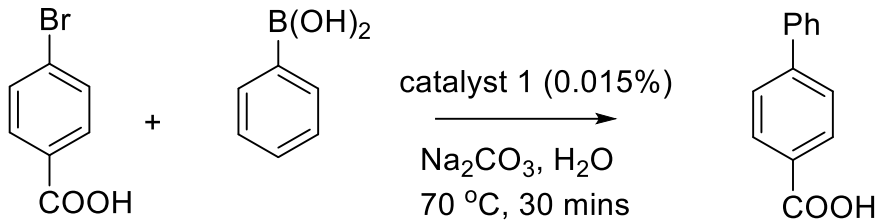
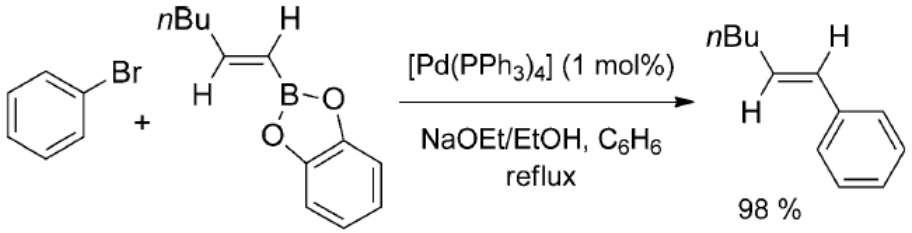
CM3191 Org L1
Suzuki-Miyaura Cross-Coupling in Water



Dr. Hoang Truong Giang
chmhoan@nus.edu.sg

Suzuki-Miyaura cross-coupling (or Suzuki coupling)

- The palladium-catalyzed cross-coupling reaction between an organoboron compound (boronic acid or ester) and a vinyl/aryl halide or triflate



Suzuki and Miyaura 1979

- General reaction:

$$R^1\text{-}B(R)_2 \quad + \quad R^2\text{-}X \xrightarrow[\text{base}]{\text{Pd(0)}} R^1\text{-}R^2$$

- The coupling is stereoselective and regioselective.

Nobel Prize in Chemistry 2010



© The Nobel Foundation. Photo: U. Montan
Richard F. Heck
Prize share: 1/3

© The Nobel Foundation. Photo: U. Montan
Ei-ichi Negishi
Prize share: 1/3

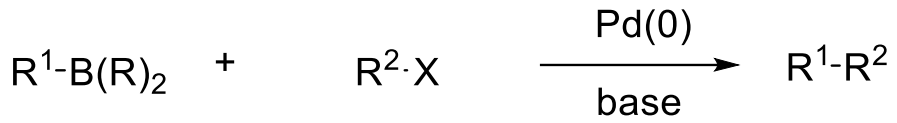
© The Nobel Foundation. Photo: U. Montan
Akira Suzuki
Prize share: 1/3

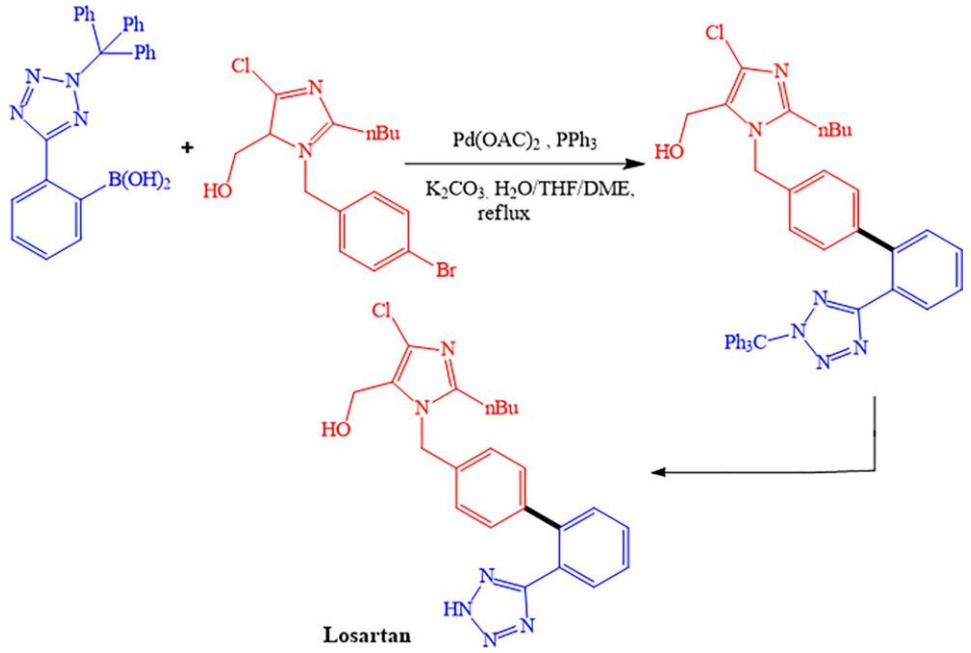
The Nobel Prize in Chemistry 2010 was awarded jointly to Richard F. Heck, Ei-ichi Negishi and Akira Suzuki "for palladium-catalyzed cross couplings in organic synthesis"

https://www.nobelprize.org/prizes/chemistry/2010/summary/

Pharmaceutical applications

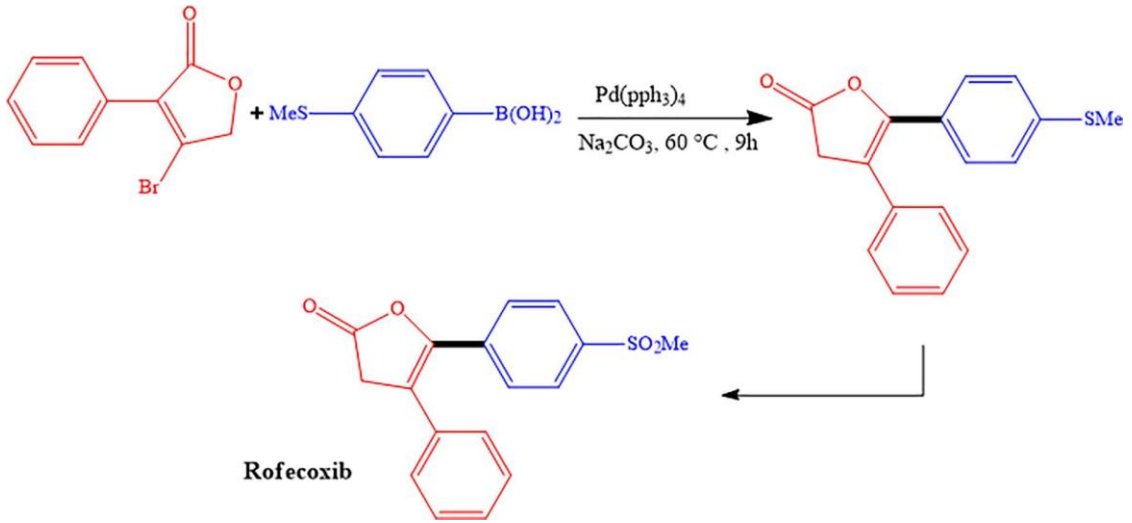
- General reaction:





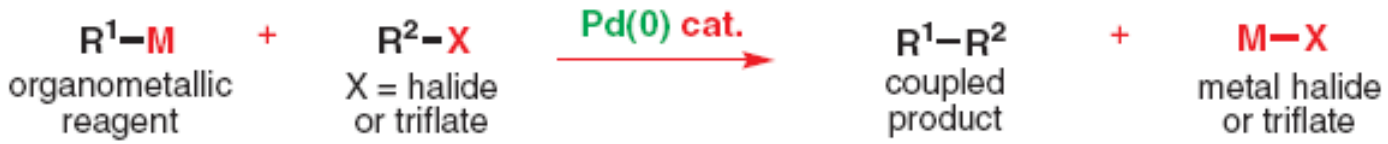
Losartan: A medication used to treat hypertension (high blood pressure).

Rofecoxib, marketed under the brand name Vioxx®, is used to relieve some symptoms caused by arthritis, such as inflammation, swelling, stiffness, and joint pain



Cross-coupling reactions

- The principle of palladium-catalyzed cross couplings is that two molecules are assembled on the metal via the formation of metal-carbon bonds. They couple to one another, and this leads to the formation of a new carbon-carbon single bond

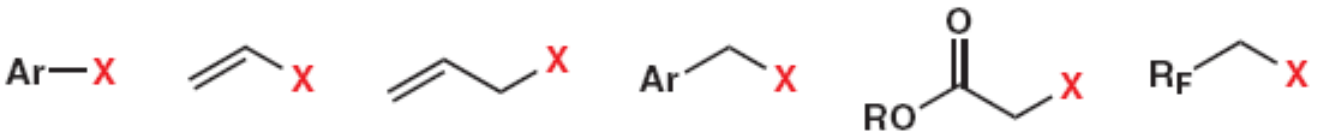


R^1–M R^1 = almost anything including examples with β H

M = MgX, ZnX, Cu, SnR_3, SiR_3, $ZrCp_2Cl$, $AlMe_2$, $B(OR)_2$, BF_4^-

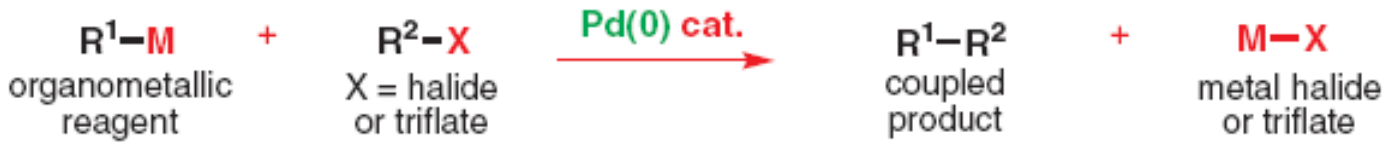
Cp = cyclo-pentadienyl

R^2–X R^2 must not have β Hs that can eliminate X = I, Br, (Cl), OTf, $OPO(OR)_2$



R_F = per-fluoroalkyl

General mechanism

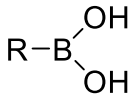


1. Oxidative addition: the oxidation state of metal is increased by two (Pd(0) → Pd(II)) and the metal gains 2 ligands

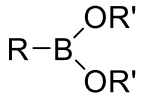
2. Transmetallation: involves the transfer of ligands from one metal to another

3. Reductive elimination: the oxidation state of metal decreases by two (Pd(II) → Pd(0)). A new σ bond is formed between the two ex-ligands.

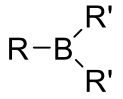
Organoboron compounds



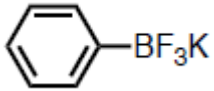
boronic acid



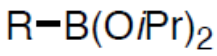
boronate ester

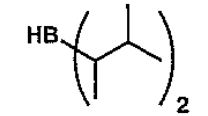


borane

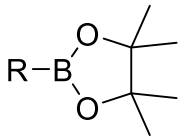


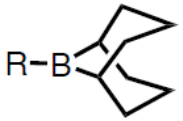
aryltrifluoroborates



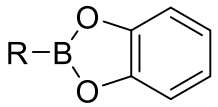


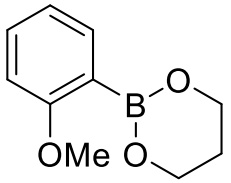
disiamylborane





9-BBN borane





For more information: Lloyd-Jones; Selection of boron reagents for Suzuki–Miyaura Coupling ***Chem. Soc. Rev.,*** 2014, 43, 412

Advantages of organoboron compounds

High Functional Tolerance:

Boronic acids tolerate diverse reactive functional groups including aldehydes, ketones, esters, and free alcohols without requiring complex protecting-group sequences.

Benign Environmental Profile:

Unlike organostannanes (Stille, highly toxic tin residues) or organozincs (Negishi, pyrophoric), organoborons produce non-toxic boric acid salts easily washed into aqueous waste

Air & Moisture Stability:

Boronic acids, pinacol boronates, and potassium trifluoroborates ($R–BF_3K$) are bench-stable crystalline solids that can be weighed on open balances and stored indefinitely.

it is suitable for industrial processes

“The Suzuki (or Suzuki-Miyaura) cross-coupling reaction is the single most numerous reaction within the C-C bond forming group, accounting for 40% of all such reactions. This popularity is almost certainly due to the almost unique combination of reagent stability and safety.

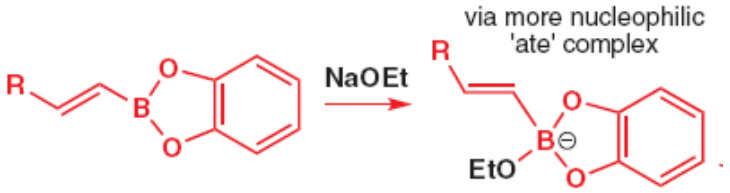
Alternative Pd-catalyzed processes are generally only employed when issues of stability or reactivity prevent a successful outcome in a Suzuki reaction”

The Medicinal Chemist’s Toolbox: An Analysis of Reactions Used in the Pursuit of Drug Candidates by by Stephen D. Roughley and Allan M. Jordan. *J. Med. Chem*. **2011**, 54, 3451–3479.

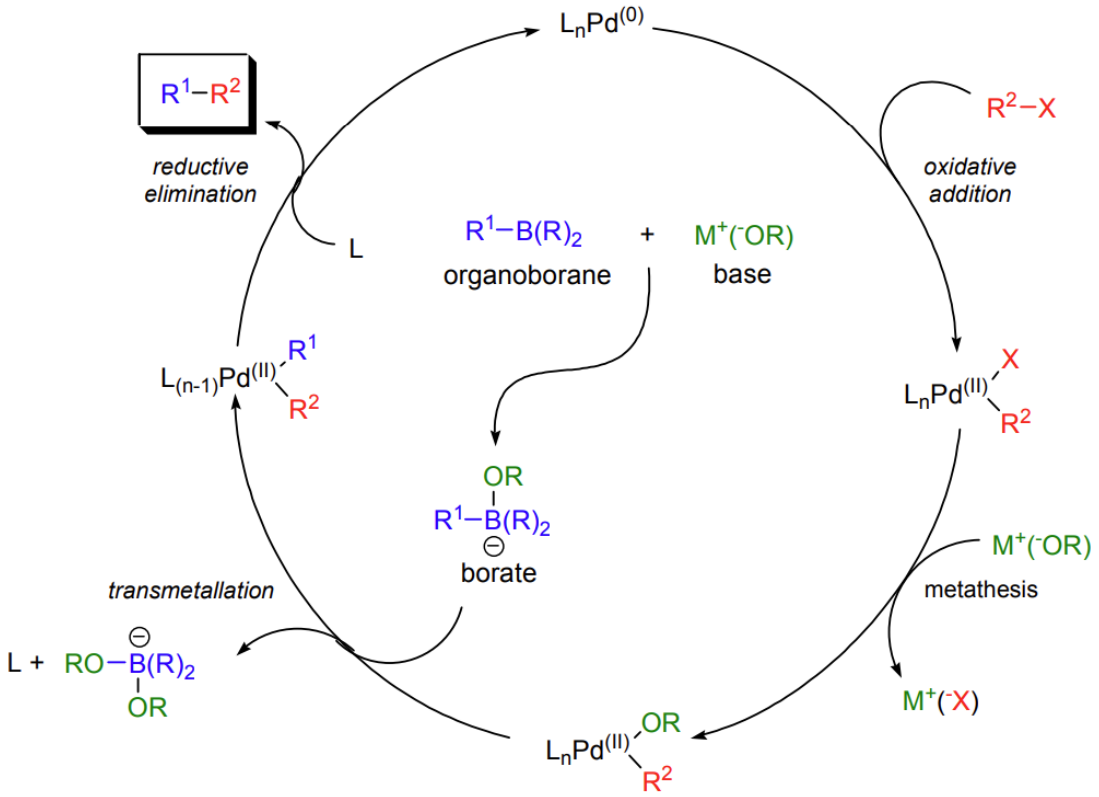
Mechanism: role of base

$R^1\text{-}B(R)_2 + R^2\text{-}X \xrightarrow[\text{base}]{Pd(0)} R^1\text{-}R^2$

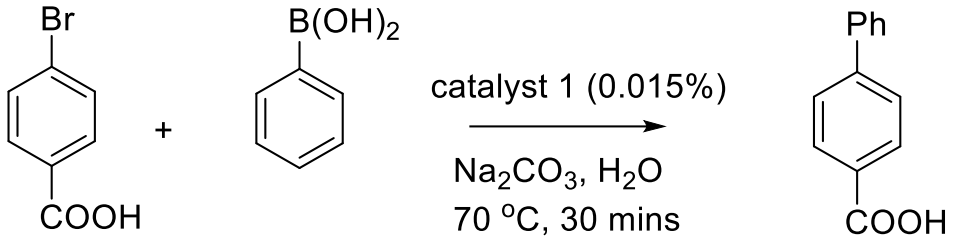
- Organoboron compounds are highly covalent in character and do not undergo transmetallation readily.
- Boronate (or “ate”) complexes, formed via quaternization of the boron with a negatively charged base are frequently invoked.

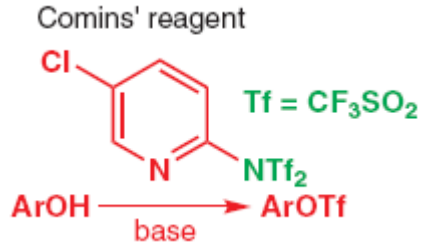


- Common bases: Na_2CO_3, $Ba(OH)_2$, K_3PO_4, TlOH, NaOH,



Experiment 1





Triflate: OTf

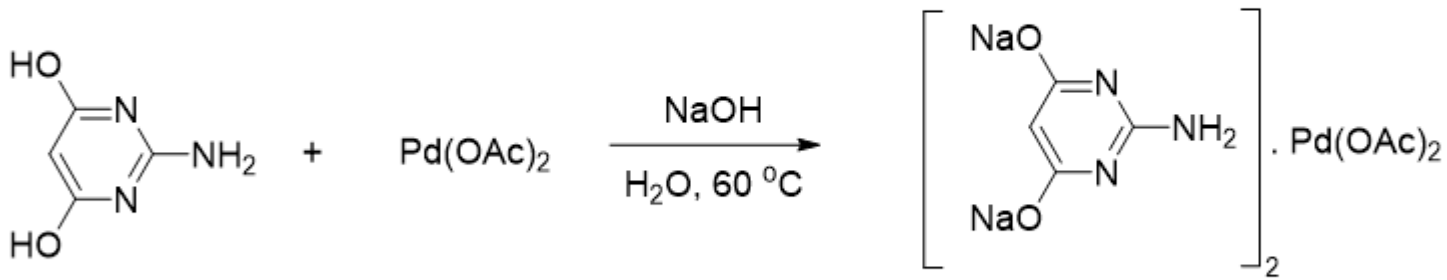
Halide substrate

```
Aryl-I  >  Aryl-Br  ≈  Aryl-OTf  ≫  Aryl-Cl  ≫
                                    Aryl-F
```

Solvent:

1. What are the key advantages and challenges of using water as a reaction solvent?
2. Why does this specific reaction succeed in water?

Appendix 1: Preparation of palladium catalyst *(by staff)*

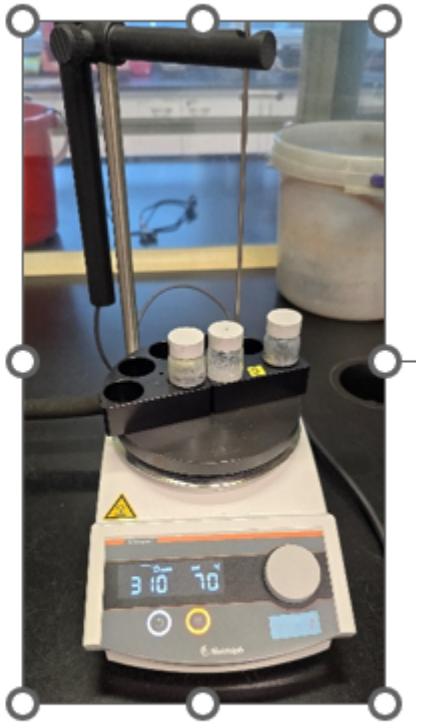


1

Add palladium acetate (5.6 mg, 0.025 mmol) and 2-amino-4,6-dihydroxypyrimidine (6.3 mg, 0.050 mmol) to a 50 mL beaker. In a separate beaker, add sodium hydroxide (4.0 mg, 0.10 mmol) to 10 mL of deionised water and stir until dissolved. Transfer the solution of sodium hydroxide into the first beaker containing the palladium acetate and the ligand. Heat the stirred mixture in a 60°C water bath for 20 minutes or until all materials have dissolved. The solution should appear yellow-orange and have no solid material. Finally, transfer the catalyst solution into a 100 mL volumetric flask and fill to line with deionised water. This concentration of palladium in this catalyst solution is 0.25 mM and should appear pale yellow.

Procedure

3.1. Reaction setup



- Turn on the hotplate stirrer equipped with heating blocks and set the temperature to 70 °C (Note: this is a shared equipment).
- Add 0.34 g of 4-bromobenzoic acid, 0.25 g of phenylboronic acid and a stir bar to a 20 mL reaction vial.
- Add 10 mL Na_2CO_3 solution (0.5 M) to the reaction vial, close the cap and stir the resulting mixture vigorously, using the hotplate stirrer in your assigned fumehood (not the common hotplate at 70 ºC), until all the reactants are dissolved.
- Add 1 mL of the palladium catalyst solution (0.25 mM) to the reaction vial, close the cap and place the vial in the preheated heating block at 70 °C. Allow the reaction to run for 30 minutes.

Why reaction vial?

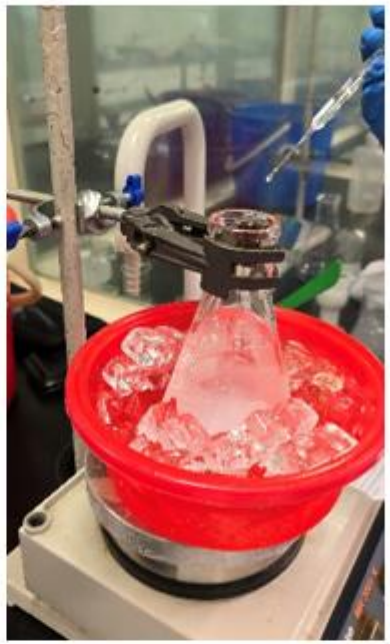
Reaction vials vs round bottom flask

Feature	Reaction Vial	Round-Bottom Flask (RBF)
Typical Scale	0.5 mL to 40 mL (milligram scale)	
Closure & Seal	Screw caps or crimp caps with PTFE/silicone septa	Ground glass standard taper joints (e.g., 14/20, 24/40)
Pressure Handling	High pressure tolerance (especially thick-walled microwave/pressure vials)	
Vacuum Handling	Limited (thin septa may deform; small neck)	Excellent for rotary evaporation and high-vacuum lines (Schlenk line)
Heating Method	Heating blocks, aluminum inserts, microwave cavities	Oil baths, heating mantles, sand baths
Common Uses		

Procedure

3.2. Reaction workup

- Take the reaction mixture off the hotplate and cool it to room temperature. Transfer the mixture into a 250 mL conical flask and cool the conical flask in an ice bath, with stirring.





- While the mixture is stirring, slowly add 20 mL of 1M HCl into the flask dropwise (note: gas will form). Check that the mixture is acidic with pH paper. Add 20 mL of water and continue to stir the mixture for 10 minutes in the ice bath.
- Isolate the crude product using suction filtration.

$$Ar\text{-}COO^{-}\ Na^{+} + HCl \rightarrow Ar\text{-}COOH\ (solid)\downarrow + NaCl$$

Procedure

3.3. Purification:

- Transfer the crude product into a 50 mL conical flask (or a 50 mL beaker) with a stir bar. Add 3 mL of 1M HCl and 10 mL of ethanol to the mixture (solution 1).
- In another 50 mL conical flask, add 20 mL of ethanol and a few boiling chips (solution 2).
- Heat the two solutions to gentle boiling. Using a dropper, slowly add just enough warm ethanol from solution 2 to solution 1 to dissolve all the material. Note: you should use a minimal amount of ethanol to dissolve the solid.



- Cool the solution to room temperature, undisturbed (about 10-15 mins). Then, place both solutions 1 and 2 in an ice bath for 15 minutes to complete the recrystallisation.
- Isolate the purified product using suction filtration. Use a minimal amount of cold ethanol (solution 2) to transfer and wash the product, if necessary.
- Transfer the crystals to a large filter paper labelled with your name, and leave it to dry under an IR lamp for 15 mins.

Procedure

3.4. Characterization

- Transfer the product into a pre-weighed plastic bag. Weigh and record the mass of the purified product. Calculate the percentage yield.
- Obtain an IR spectrum of the product.
- Prepare a sample for NMR analysis: weigh about 10 mg of the sample in a clean and dry plastic vial. Dissolve it in 0.6 mL of the NMR solvent and transfer the solution into an NMR tube using a clean and dry glass dropper. Ensure the height of the solution in the NMR tube is about 5 cm and top it up with the same NMR solvent if needed. Cap and label the NMR tube.

Turnover Number (TON) and Turnover Frequency (TOF)

Turnover Number (TON)
TON represents the total number of catalytic cycles a catalyst can complete before becoming deactivated or before the reaction is stopped. It is a dimensionless measure of catalyst stability and longevity.

$$\text{TON} = \frac{\text{Moles of product formed}}{\text{Moles of active catalyst}}$$

A higher TON indicates that a smaller amount of catalyst is required to convert a large quantity of substrate, signifying a robust and durable catalyst.

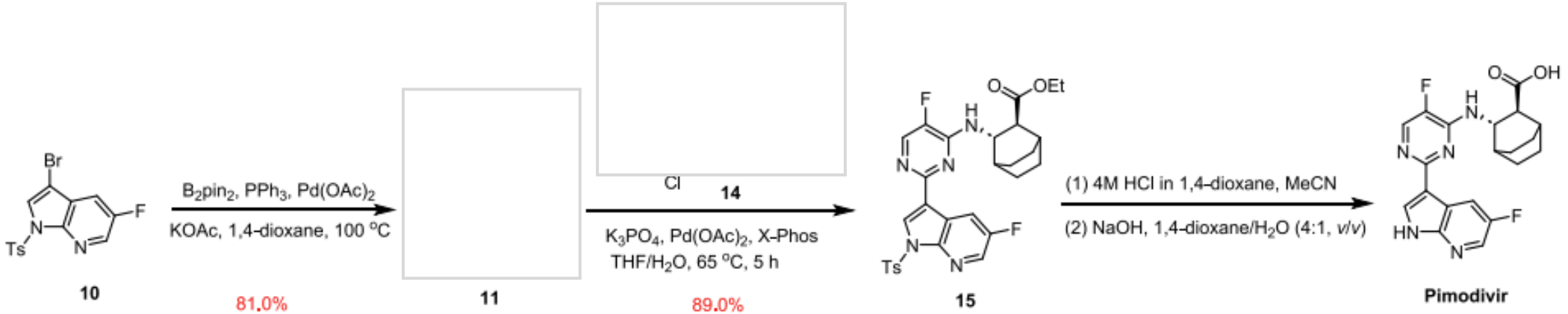
Turnover Frequency (TOF)
TOF measures the rate of catalytic turnover per active catalytic site per unit time. It quantifies catalytic activity and speed. **Units:** Typically expressed in s^{-1}, min^{-1}, or h^{-1}.

$$\text{TOF} = \frac{\text{TON}}{\text{Time}} = \frac{\text{Moles of product formed}}{\text{Moles of active catalyst} \times \text{Time}}$$

A higher TOF indicates a faster catalytic cycle (a higher intrinsic rate of reaction at each catalytic center).

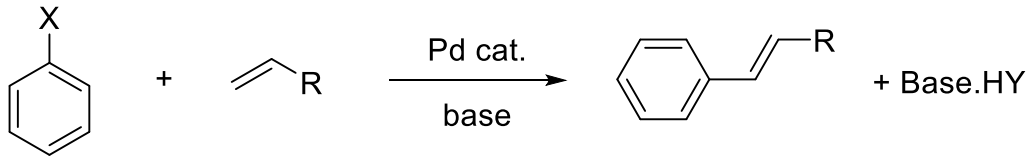
Exercise

Provide the structure of **11** (organoboron compound) and **14** (a chloride compound) for the synthesis of pimodivir, an antiviral drug which was developed as a treatment for influenza

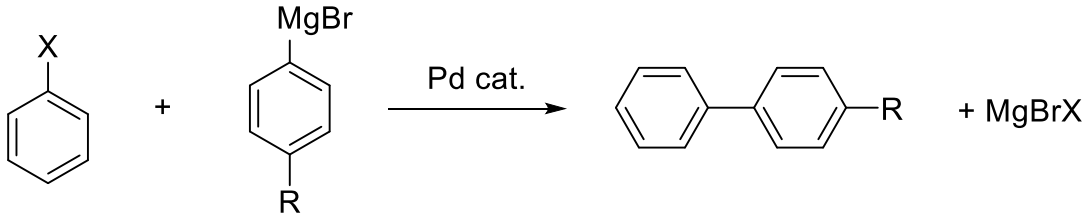


Appendix: other coupling reactions

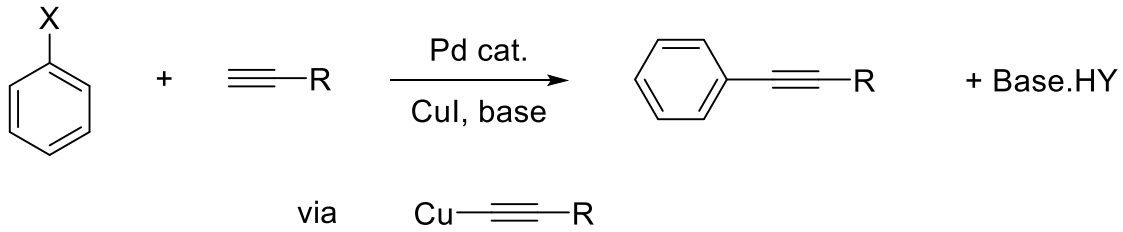
Mizoroki - Heck coupling (1968-1972)

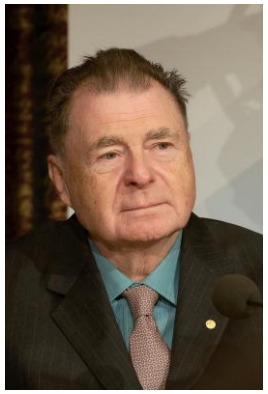


Kumada coupling (1972 Nickel catalyst)



Sonogashira coupling (1975)



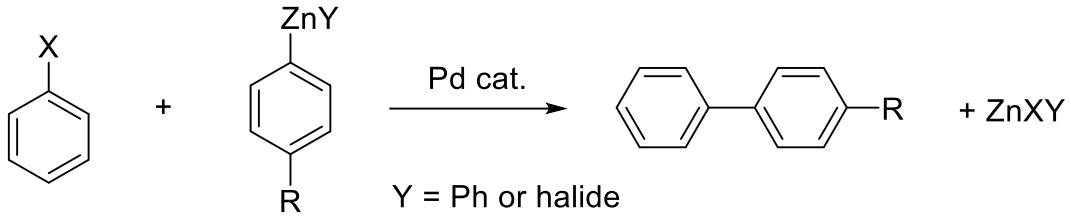


Prof Richard Frederick Heck
(1931-2015)
University of Delaware

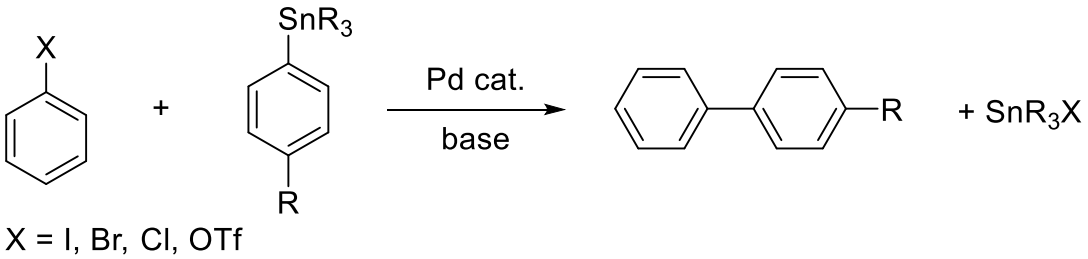
https://en.wikipedia.org/wiki/Richard_F._Heck

Other coupling reactions

Negishi coupling (1976-1977)



Stille coupling (1976-1978)





Prof Ei-ichi Negishi
(1935-2021)
Purdue University

https://en.wikipedia.org/wiki/Ei-ichi_Negishi

Other coupling reactions

- 1988-1994 Hiyama reaction: organosilicon
- 1995 Buchwald - Hartwig coupling: C-N and C-O formation

